Congruences for the series A with $A\big(x - x\,A(4x)/4\big) = x$

A proof of two conjectures in OEIS A393856

Dev D. Goyal

October 2026

Abstract

The sequence A393856 of the On-Line Encyclopedia of Integer Sequences (OEIS) lists the coefficients of the power series $A(x) = x + x^2 + 6x^3 + 121x^4 + \cdots$ defined by the equation $A\big(x - x\,A(4x)/4\big) = x$. The unknown series A appears on both sides of this equation, and its coefficients grow very fast. Paul D. Hanna conjectured that every coefficient leaves remainder 1 on division by 5, and that a coefficient is odd exactly when its index is a power of 2. We prove both conjectures. In fact we prove two statements that hold with 4 replaced by any positive integer q: modulo $q+1$ the series becomes $x + x^2 + x^3 + \cdots$, and, when q is even, modulo 2 it agrees with the generating series of the Catalan numbers, whose odd coefficients sit exactly at the powers of 2. The key observation is that in arithmetic modulo $q+1$ the number q behaves like -1, and for $q = -1$ the equation can be solved by hand. The results have also been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [3] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A393856 [1], contributed by Paul D. Hanna in March 2026. It defines a power series

$$A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$$

by the equation

$$A(x - x\,A(4x)/4) = x. \tag{1}$$

Here $A(4x)$ is the series A with x replaced by $4x$, so its coefficient of x^n is $4^n\,a(n)$. Dividing by 4 gives the series whose coefficients are $4^{\,n-1}a(n)$, which are still integers. The left side of (1) means: substitute the series $x - x\,A(4x)/4$ into A. (Section 2 explains what substituting one series into another means.) So the equation says that A undoes a substitution that is itself built from A. Because A appears on both sides, it is not obvious that the equation has a solution, or only one; Theorem 3.1 shows that it has exactly one solution of the form $x +$ (higher powers of x).

The first terms, starting at $n = 1$, are 1, 1, 6, 121, 8466, 2227706, 2299117036, ... They grow very fast: $a(15)$ already has 56 digits. The entry reads:

```
G.f.  A(x) satisfies:  A( x - x*A(4*x)/4 ) = x.

Conjecture:  a(n) is odd iff n is a power of 2.

Conjecture:  a(n) == 1 (mod 5) for n >= 1.
```

— OEIS A393856, Paul D. Hanna, March 2026

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence. The entry also records other formulas for $A(x)$, for example as an infinitely nested sequence of substitutions; we do not need them.

1.2 The question

Hanna conjectured, from the computed terms, that every $a(n)$ leaves remainder 1 on division by 5, and that $a(n)$ is odd exactly when n is a power of 2, that is, when n is one of $1, 2, 4, 8, 16, \ldots$. The table shows the remainders of the first 32 terms modulo 5 and modulo 2. The indices that are powers of 2 are printed in bold.

n	**1**	**2**	3	**4**	5	6	7	**8**	9	10	11	12	13	14	15	**16**
$a(n)$ mod 5	**1**	**1**	1	**1**	1	1	1	**1**	1	1	1	1	1	1	1	**1**
$a(n)$ mod 2	**1**	**1**	0	**1**	0	0	0	**1**	0	0	0	0	0	0	0	**1**

n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	**32**
$a(n)$ mod 5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	**1**
$a(n)$ mod 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	**1**

1.3 What this paper proves

Main result (Corollary 3.4). *Both conjectures are true:* $a(n) \equiv 1 \pmod 5$ *for every* $n \geq 1$, *and* $a(n)$ *is odd if and only if* n *is a power of* 2.

The number 4 plays no special role. Hanna contributed a whole family of such sequences, in which 4 is replaced by another positive integer q, and we treat all of them at once. Theorem 3.2 shows that, for every $q \geq 1$, all coefficients leave remainder 1 on division by $q+1$. Theorem 3.3 shows that, for every even q, a coefficient is odd exactly when its index is a power of 2.

The key observation is very simple. In arithmetic modulo $q + 1$, the number q is the same as -1, and when $q = -1$ the equation has the obvious solution $x + x^2 + x^3 + \cdots$. In the same way, modulo 2 an even number q is the same as 0, and when $q = 0$ the equation becomes the classical equation of the Catalan numbers, whose parity is well known.

1.4 What was known

Hanna contributed the four entries A393856–A393859, for $q = 4, 5, 6, 7$, in March 2026, each with the conjectures for its own value of q. The OEIS also lists the members for $q = 0, 1, 2, 3$, which were contributed earlier. For $q = 0$, where $A(qx)/q$ is replaced by x (its value at $q = 0$ according to formula (2) below), the equation becomes $A(x - x^2) = x$; its solution is the generating series of the Catalan numbers (A000108). The cases $q = 1, 2, 3$ are A088714, A195195 and A195196. We are not aware of an earlier proof of the conjectures. The tools we use, formal power series and their reduction modulo an integer, are standard; see [5, 4].

1.5 Why it is interesting

Equations in which an unknown series is substituted into an expression built from itself are common in combinatorics, where they describe structures defined recursively, such as trees. The Catalan numbers, the case $q = 0$ of this family, are the classic example. Hanna has contributed many such equations to the OEIS. For $q \geq 1$ the coefficients grow so fast that no direct formula for them is in sight, so it is striking that their remainders follow such simple rules. The proof explains the rules: working modulo $q + 1$ or modulo 2 replaces the parameter q by -1 or by 0, two values at which the equation can be solved by hand. This idea, choosing a modulus in which the equation becomes simple, runs through this whole collection of papers.

1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution, and remainders. Section 3 makes the equation precise for every q and states the results, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the results, a further congruence, and open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\, U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1 - x)(1 + x + x^2 + x^3 + \cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are

determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of U below degree d are zero. So $U - V = O(x^d)$ means that U and V agree up to degree $d-1$. A series of the form $x + u_2x^2 + u_3x^3 + \cdots$, with constant term 0 and coefficient of x equal to 1, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \ge 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 5 \pmod 4$, since both leave remainder 1, and $-3 \equiv 1 \pmod 4$. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod 2$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Several lemmas below hold over every *commutative ring*, that is, every set with an addition, subtraction and multiplication satisfying the usual rules of arithmetic (for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$). No division is used; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The method

Almost every proof in this paper follows the same three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/m\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo m gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A good guess.* Find a simple series that satisfies the reduced equation and check that it does. By uniqueness, this simple series *is* the reduction of the sequence, and its coefficients can simply be read off.

The creative part is step 3: finding a simple series that works. Step 1 sometimes needs care, when the equation seems to require a division.

2.6 Squaring modulo 2

Lemma 2.2 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 just doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod{2}$. In particular, the square of any series has, modulo 2, no terms of odd degree.

2.7 The Catalan numbers modulo 2

Lemma 2.3. *In every commutative ring the equation $U = x+U^2$ has exactly one solution with $U(0) = 0$. Over the integers it is the Catalan series $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, and*

$$C(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k\geq 0} x^{2^k} \pmod 2.$$

So $[x^n]\,C$ is odd exactly when n is a power of 2.

Proof. The coefficient of x^n in U^2 involves only $u_1, \ldots, u_{n-1}$, so the equation determines u_n one at a time: $u_1 = 1$ and $u_n = \sum_{i=1}^{n-1} u_i u_{n-i}$. Over $\mathbb{F}_2$, the series $S = \sum_k x^{2^k}$ satisfies $x + S^2 = x + \sum_k x^{2^{k+1}} = S$ by Lemma 2.2, so by uniqueness $C \equiv S \pmod 2$. □

The parity of the Catalan numbers is classical; see for example [2]. (Here the Catalan number $\frac{1}{n}\binom{2n-2}{n-1}$ appears as the coefficient of x^n, so "n a power of 2" corresponds to the familiar statement that the k-th Catalan number is odd exactly when $k+1$ is a power of 2.)

3 Results

To treat the whole family at once, we replace 4 by a parameter q. First we make the equation precise. For a series $U = u_1x + u_2x^2 + u_3x^3 + \cdots$ without constant term and a number $q \neq 0$, the series $U(qx)/q$ has the coefficients $q^{n-1}u_n$. We take this formula as the definition, for every q:

$$L_q(U) = \sum_{n\geq 1} q^{\,n-1} u_n\, x^n = u_1x + q\,u_2x^2 + q^2u_3x^3 + \cdots . \tag{2}$$

No division by q occurs, so $L_q(U)$ makes sense for every q, including $q = 0$ (where it equals u_1x), and for coefficients in every commutative ring, such as $\mathbb{Z}/m\mathbb{Z}$. The equation of the family is

$$A(x - x\,L_q(A)) = x, \qquad A \text{ normalized.} \tag{3}$$

For $q = 4$ this is equation (1) of the entry.

Theorem 3.1 (Exactly one solution)**.** *Let q be an element of a commutative ring R. There is exactly one normalized series G with coefficients in R that satisfies* (3)*. When $R = \mathbb{Z}$ and q is an integer, we call this solution G_q. For every $m \geq 2$, the reduction of G_q modulo m is the solution of* (3) *over $\mathbb{Z}/m\mathbb{Z}$, with q replaced by its residue.*

So the sequence of OEIS A393856 is $a(n) = [x^n]\,G_4$.

Theorem 3.2 (All coefficients are 1 modulo $q+1$)**.** *For every integer $q \geq 1$,*

$$G_q(x) \equiv \frac{x}{1-x} = x + x^2 + x^3 + \cdots \pmod{q+1}.$$

That is, every coefficient $[x^n]\,G_q$ with $n \geq 1$ leaves remainder 1 on division by $q+1$.

Theorem 3.3 (Parity)**.** *If q is even, then*

$$G_q(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k\geq 0} x^{2^k} \pmod 2,$$

so $[x^n]\,G_q$ is odd if and only if n is a power of 2. If q is odd, every $[x^n]\,G_q$ with $n \geq 1$ is odd.

Corollary 3.4 (Hanna's conjectures)**.** *For the sequence of OEIS A393856: $a(n) \equiv 1 \pmod 5$ for every $n \geq 1$; and $a(n)$ is odd if and only if n is a power of 2.*

4 Proofs

The idea in brief. The equation determines the coefficients of A one at a time, without any division, so it has exactly one solution over every commutative ring, including $\mathbb{Z}/(q+1)\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ (Theorem 3.1). Modulo $q+1$ we may replace q by -1, and then $x + x^2 + x^3 + \cdots$ is easily seen to be a solution. Modulo 2, for even q, we may replace q by 0, and then the equation says that A undoes the substitution $x \mapsto x - x^2$, which is exactly what the Catalan series does. By uniqueness, these simple series are the reductions of G_q.

Proof of Theorem 3.1. Write $G = x + g_2x^2 + g_3x^3 + \cdots$ with unknown coefficients, and put $V = x - x\,L_q(G)$. By (2), the coefficient g_k enters $L_q(G)$ only in degree k, and therefore enters V only in degree $k+1$, because of the factor x. Also, $V = x + O(x^2)$ is normalized. Now look at the coefficient of x^n, for some $n \geq 2$, in

$$G(V) = V + g_2V^2 + g_3V^3 + \cdots + g_nV^n + \cdots .$$

- The term g_nV^n contributes exactly g_n, because $V^n = x^n + O(x^{n+1})$.
- The terms g_kV^k with $k > n$ contribute nothing, because V^k has no terms below x^k.
- The remaining terms, V and g_kV^k with $2 \leq k < n$, involve g_k with $k < n$, and the coefficients of V up to degree n, which involve only $g_2, \ldots, g_{n-1}$.

So the coefficient of x^n in $G(V)$ is g_n plus an expression in $q, g_2, \ldots, g_{n-1}$ built with additions, subtractions and multiplications. Setting it equal to the coefficient of x^n in x, which is 0, determines g_n. Doing this for $n = 2, 3, 4, \ldots$ in turn gives exactly one solution, in every commutative ring.

Reduction modulo m is compatible with all the operations in (3): with L_q, which multiplies the coefficients by powers of q, with products, and with substitution. So the reduction of G_q satisfies the reduced equation, and by the uniqueness just proved (over $\mathbb{Z}/m\mathbb{Z}$) it is its solution. □

Proof of Theorem 3.2. We work over $\mathbb{Z}/(q+1)\mathbb{Z}$, where $q = -1$. Let $H = x/(1-x) = x + x^2 + x^3 + \cdots$. All coefficients of H (except the constant term) are 1, so by (2)

$$L_{-1}(H) = x - x^2 + x^3 - x^4 + \cdots = \frac{x}{1+x}, \qquad \text{hence} \qquad x - x\,L_{-1}(H) = x - \frac{x^2}{1+x} = \frac{x}{1+x}.$$

Substituting this into H gives

$$H\Big(\frac{x}{1+x}\Big) = \frac{x/(1+x)}{1 - x/(1+x)} = \frac{x/(1+x)}{1/(1+x)} = x.$$

So H satisfies the reduced equation. By Theorem 3.1, H is the reduction of G_q modulo $q+1$. Every coefficient of H of positive degree is 1. □

Proof of Theorem 3.3. First let q be even, and work over $\mathbb{F}_2$, where $q = 0$. By (2), $L_0(U) = u_1x$, because all other coefficients carry a factor $0^{n-1} = 0$. For a normalized U this is just x, so over $\mathbb{F}_2$ the equation (3) becomes

$$G(x - x^2) = x.$$

In other words, G undoes the substitution $x \mapsto x - x^2$, so G is the compositional inverse of $x - x^2$ (Lemma 2.1). Over the integers this inverse is the Catalan series $C(x) = x + x^2 + 2x^3 + 5x^4 + \cdots$ (Section 2), and reduction modulo 2 preserves the equation $C(x - x^2) = x$. By the uniqueness

in Theorem 3.1, the reduction of G_q modulo 2 equals the reduction of C. By Lemma 2.3, that is $\sum_k x^{2^k}$.

If q is odd, then $q+1$ is even. By Theorem 3.2, every coefficient of positive degree is 1 plus a multiple of $q+1$, hence odd. □

Corollary 3.4 is the case $q = 4$ of Theorems 3.2 and 3.3.

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We solved (1) coefficient by coefficient with exact integer arithmetic, up to $n = 25$; this reproduces all 15 terms printed in the OEIS entry. We then ran the same computation modulo $10 = 2(q+1)$, which is legitimate because the construction uses only additions, subtractions and multiplications (Theorem 3.1). It agrees with the exact values for $n \le 25$, and it confirms both statements of Corollary 3.4 for every $n < 300$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [8]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [6] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [7], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A393856.lean` of the repository, in the namespace `OEIS.A393856`. In Lean, the operation L_q of (2) is called `scaleTail q`. The general theorems for every q are `integer_mod_parameter_add_one` (Theorem 3.2) and `integer_coeff_odd_iff` (Theorem 3.3 for even q), in the namespace `Hanna.Related.ScaledInverse`, in the files `lean/Hanna/Related/ScaledInverse.lean` and `ScaledParity.lean` of the same folder. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
Theorem 3.1, over every commutative ring	`source_existence_uniqueness`
G_4 is normalized and satisfies (1)	`source_normalized`, `source_equation`
Corollary 3.4: $a(n) \equiv 1 \pmod 5$ for $n \ge 1$	`a393856_mod_five`
Corollary 3.4: $a(n)$ is odd $\iff$ n is a power of 2 (for all $n \ge 0$)	`a393856_parity`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Corollary 3.4 settles the conjectures of the entry completely, for every index n. The theorems describe the remainders modulo $q+1$ and modulo 2; they say nothing about the size of the coefficients.

A further congruence. The argument used for Theorem 3.3 works modulo q itself. In arithmetic modulo q the parameter q is 0, so the equation becomes $G(x-x^2)=x$ again, and G_q agrees modulo q with the Catalan series:

$$[x^n]\,G_q \equiv \frac{1}{n}\binom{2n-2}{n-1} \pmod{q} \qquad (n \geq 1).$$

For example, for $q=4$ and $n=5$, the term $a(5)=8466$ and the Catalan number $\frac{1}{5}\binom{8}{4}=14$ both leave remainder 2 on division by 4. Since q and $q+1$ have no common factor, a number is determined modulo $q(q+1)$ by its remainders modulo q and modulo $q+1$ (the Chinese remainder theorem). So, together with Theorem 3.2, this determines every coefficient of G_q modulo $q(q+1)=20$ when $q=4$. For even q it also contains Theorem 3.3, because 2 divides q. We checked this congruence by computer for $q=2,3,\ldots,7$ and every $n<300$. It is not part of the Lean formalization.

The rest of the family. The theorems hold for every $q \geq 1$. The OEIS lists the members $q=1,2,3$ as A088714, A195195 and A195196; we checked that their published terms agree with the coefficients of G_1, G_2 and G_3 (for A088714 after shifting the index by one). For instance, A195195 is defined by the case $q=2$ of (1), so our results show that all its terms leave remainder 1 on division by 3, and that a term is odd exactly when its index is a power of 2.

Open questions. We have not studied the remainders modulo other numbers, such as $25=(q+1)^2$ for $q=4$. It would be interesting to know whether they are also governed by a simple series.

About this paper

How the work was done. This paper is one of 26 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [8]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every

claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [8], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and the conjectures are due to Paul D. Hanna [1], who contributed the whole family A393856–A393859 ($q = 4, 5, 6, 7$), as well as the earlier members A088714, A195195 and A195196.

References

[1] P. D. Hanna, entry A393856 (G.f. $A(x)$ satisfies $A\big(x - x\,A(4x)/4\big) = x$), *The On-Line Encyclopedia of Integer Sequences*, March 2026. `https://oeis.org/A393856`. Text as of October 2026.

[2] E. Deutsch and B. E. Sagan, Congruences for Catalan and Motzkin numbers and related sequences, *Journal of Number Theory* 117 (2006), 191–215.

[3] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[4] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[5] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[6] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[7] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `c44e0c8ee63c`.

[8] D. D. Goyal, *Proofs and corrections of OEIS conjectures, with Lean 4 formalizations*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.